Engineering Probability
Homework 1

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(1)

An integrated circuit factory has three machines X, Y and Z. Test one integrated circuit produced by each machine. Either a circuit is acceptable (a) or it fails (f). An observation is a sequence of three test results corresponding to the circuits from machines X, Y and Z respectively. For example aaf is the observation that the circuits from X and Y pass the test and the circuit from Z fails the test.

a What are the elements of the sample space of this experiment?

$S = \{aaa, aaf, afa, aff, faa, faf, ffa, fff\}$

b What are the elements of the sets:

$$Z_F = \{\text{circuit from } Z \text{ fails}\} = \{aaf, aff, faf, fff\}$$
$$X_A = \{\text{circuit from } X \text{ is acceptable}\} = \{aaa, aaf, afa, aff\}$$

c Are Z_F and X_A mutually exclusive? No the element aaf is in both sets.

d Are Z_F and X_A collectively exhaustive?

No the element ffa is not included in either of the sets.

e What are the elements of the sets:

$$C = \{\text{more than one circuit acceptable}\} = \{aaa, aaf, afa, faa\}$$
$$D = \{\text{at least two circuits fail}\} = \{aff, faf, ffa, fff\}$$

f Are C and D mutually exclusive?

Yes

g Are C and D collectively exhaustive?

Yes

(2)

There are two types of cellular phones, hand-held phones (H) that you carry and mobile phones (M) that are mounted in vehicles. Phone calls can be classified by the traveling speed of a user as fast (F) or slow (W). Monitor a cellular phone call and observe the type of telephone and speed of the user. The probability model for this experiment has the following information: P[F] = 0.5, P[HF] = 0.2, P[MW] = 0.1. What is the sample space of the experiment? Calculate the following probabilities:

$S = \{HF, HW, MF, MW\}$

a $P[W] = P(F^c) = 1 - P(F) = 0.5$

b $P[MF]$, $P(F) = P(HF) + P(MF)$, $\quad P(MF) = P(F) - P(HF) = 0.5 - 0.2 = 0.3$

c $P[H]$= $P(M^c) = 1 - (P(MF) + P(MW)) = 1 - 0.3 - 0.1 = 0.6$

(3)

Cellular phones perform handoffs as they move from cell to cell. During a phone call, telephone either performs zero handoffs (H_0), one handoff (H_1), or more than one handoff (H_2). In addition, each call is either long (L), if it lasts more than 3 minutes, or brief (B). The following table describes the possible types of calls.

	H_0	H_1	H_2
L	0.1	0.1	0.2
B	0.4	0.1	0.1

What is the probability $P[H_0]$ that a phone makes no handoffs? What is the probability a call is brief? What is the probability that the call is long or there are at least two handoffs?

$P[H_0] = P[H_0L] + P[H_0B] = 0.4 + 0.1 = 0.5$

$P[B] = P[H_0B] + P[H_1B] + P[H_2B] = 0.4 + 0.1 + 0.1 = 0.6$

$P[L \cup H_2] = P[L] + P[H_2] - P[L \cap H_2] = 0.4 + 0.3 - 0.2 = 0.5$

(4)

Monitor three consecutive phone calls going through a telephone switching office. Classify each one as a voice call (v), if someone is speaking or a data call (d) if the call is carrying a modem or fax signal. Your observation is a sequence of three letters (*each letter is either* v *or* d). For example two voice calls followed by one data call corresponds to vvd. Write the elements of the following sets: Write the elements of:

a. $A_1 = \{\text{first call is voice}\} = \{vvv, vvd, vdv, vdd\}$

b. $B_1 = \{\text{first call is data}\} = \{dvv, dvd, ddv, ddd\}$

c. $A_2 = \{\text{second call is voice}\} = \{vvv, vvd, dvv, dvd\}$

d. $B_2 = \{\text{second call is data}\} = \{vdv, vdd, ddv, ddd\}$

e. $A_3 = \{\text{all calls the same}\} = \{vvv, ddd\}$

f. $B_3 = \{\text{voice and data alternate}\} = \{vdv, dvd\}$

g. $A_4 = \{\text{one or more voice calls}\} = \{vvv, vvd, vdv, vdd, dvv, dvd, ddv\}$

h. $B_4 = \{\text{two or more data calls}\} = \{vdd, dvd, ddv, ddd\}$

For each pair of events A_1 and B_1, A_2 and B_2 and so on, please identify whether the pair of events is either mutually exclusive or collectively exhaustive

A_1 and B_1 are collectively exhaustive and mutually exclusive.
A_2 and B_2 are collectively exhaustive and mutually exclusive.
A_3 and B_3 are mutually exclusive.
A_4 and B_4 are collectively exhaustive.

(5)

A student's test score T is an integer between 0 and 100 corresponding to the experimental outcomes $s0, \ldots, s100$. A score of 90 to 100 is an A, 80 to 89 is a B, 70 to 79 is a C, 60 to 69 is a D, and below 60 is a failing grade of F. Given all scores between 51 and 100 are equally likely and a score of 50 or less cannot occur, find the following probabilities.

Since all scores are equally likely, $P(Event) = \frac{\#offavorableoutcomes}{Total\#ofoutcomes}$

a $P[\{s_{79}\}] = \frac{1}{50} = 0.02$

b $P[\{s_{100}\}] = \frac{1}{50} = 0.02$

c $P[A] = \frac{11}{50} = 0.22$

d $P[F] = \frac{9}{50} = 0.18$

e $P[T \geq 80] = \frac{21}{50} = 0.42$

f $P[T < 90] = \frac{39}{50} = 0.78$

g $P[\text{C grade or better}] = \frac{31}{50} = 0.62$

h $P[\text{student passes}] = \frac{41}{50} = 0.82$

(6)

Monitor a phone call. Classify the call as a voice call (V), if someone is speaking; or a data call (D) if the call is carrying a modem or fax signal. Classify the call as long (L) if the call lasts for more than three minutes; otherwise classify the call as brief (B). Based on data collected by the telephone company, we use the following probability model: $P[V] = 0.7, P[L] = 0.6, P[VL] = 0.35$. Find the following probabilities:

a $P[DL] = P(L) - P(VL) = 0.6 - 0.35 = 0.25$

b $P[D \cup L] = P(D) + P(L) - P(D \cap L) = P(V^c) + 0.6 - 0.25 = (1 - 0.7) + 0.6 - 0.25 = 0.65$

c $P[VB] = P(V) - P(VL) = 0.7 - 0.35 = 0.35$

d $P[V \cup L] = P(V) + P(L) - P(V \cap L) = 0.7 + 0.6 - 0.35 = 0.95$

e $P[V \cup D] = P(S) = 1$ since V and D are collectively exhaustive.

f $P[LB] = 0$ since L and B are mutally exclusive.